

input

3	1	0	2	4
1	5	2	0	1
0	2	7	3	2
4	1	3	6	0
2	0	1	2	5

kernel

1	0	-1
2	0	-2
1	0	-1

output

-2

one cell of the
feature map

$$5 \times 1 + 0 \times -1 + 2 \times 2 + 3 \times -2 + 1 \times 1 + 6 \times -1 = -2$$

Nine multiplications and an addition. That is the whole operation; everything else is where you slide it to next.